Supplementary Material – Diele-Viegas et al. Political polarization and the future of environmental protection in Latin America and the Caribbean

Methods

Ideological discourse and institutional performance

To describe the institutional context in which environmental policies are implemented, we combined data from the Varieties of Democracy (V-Dem) dataset, version 16 (Coppedge, 2026), with a country-level matrix of recent presidential administrations in Latin America and the Caribbean (LACA). The matrix identified each country, chief executive, and administration's start and end years. We aligned annual V-Dem observations with presidential mandates, using the manually specified start and end years for each administration, and summarized the available years for the administrations included in the matrix.

We extracted variables describing regime type, institutional performance, and executive ideological legitimation. Regime type was characterized using V-Dem's Regimes of the World classification, which distinguishes political regimes from closed autocracies to liberal democracies.

For the institutional comparison presented in Figure 1, we examined public goods provision, law enforcement, state capacity, and civil society space. The comparison is descriptive and does not estimate the causal effects of political transitions or executive ideology. These indicators capture broad institutional conditions that may affect environmental governance. They are not direct measures of environmental regulation, enforcement, or political polarization. We characterize executive ideological discourse using V-Dem's legitimation indicators, including nationalist, socialist/communist, conservative, autonomist, and religious narratives.

For each mandate, annual observations were averaged across the years included in the analysis. Following the classification procedure applied to the figure, the mandate-level means were rounded to one decimal place, and a discourse dimension was treated as predominant when its rounded mean was at least 0.5.

A single predominant dimension was classified as a strict discourse category. For the purposes of this classification, predominant socialist/communist discourse was assigned to the Strict Left category, whereas predominant conservative discourse was assigned to the Strict Right category. When multiple dimensions reached the threshold, the administration was classified as Hybrid Left

if socialist/communist discourse was among the predominant dimensions, or Hybrid Right if conservative discourse was predominant without socialist/communist discourse. Predominant dimensions other than socialist/communist or conservative were grouped as Other Ideologies. Mandates for which no assessed dimension reached the threshold were classified as No Dominant Ideology. This threshold-based grouping is an operational classification for descriptive visualization, not an independently validated measure of left–right ideology or polarization; rounding may affect classifications close to the threshold.

For Javier Milei (Argentina), Daniel Noboa (Ecuador), Bernardo Arévalo (Guatemala), and Claudia Sheinbaum (Mexico), the V-Dem version used did not provide the consolidated legitimation data required by this procedure. Their discourse categories were therefore assigned separately on the basis of the cited political literature (Heinisch et al., 2024; Lasso and Venegas, 2025; Schwartz, 2024; Sepúlveda, 2026). For these administrations, ideological discourse categories were manually assigned using the cited literature before the final groups were generated. The final classifications should therefore be documented explicitly to ensure that the analysis can be reproduced without interactive modifications. These literature-based assignments are not directly equivalent to classifications derived from V-Dem scores and should be interpreted accordingly.

For Figure 1, public goods provision, law enforcement, and state capacity were rescaled from their original standardized ordinal scales (theoretical range 0–4) to a 0–1 interval to permit visual comparison with the civil society measure. Annual values were averaged within each administration before plotting. We displayed countries and administrations in country order and mandate chronology. These transformations support visualization across indicators but do not make the measures interchangeable or establish differences in environmental policy outcomes.

Temporal variation in burned area, climate, and institutional conditions

To describe fire dynamics in relation to climate variability from 2001 to 2023, we compared annual burned area from Collection 8 of the MapBiomas fire-scar dataset (MAPBIOMAS, 2026) with the Oceanic Niño Index (ONI). Historical ONI observations were obtained from the Climate Prediction Center of the National Oceanic and Atmospheric Administration (NOAA) through the

rsoi package in R (Albers, 2020; R Core Team, 2022). We then saved the extracted data and incorporated the annual ONI values into the visualization script.

ONI was included to provide context on El Niño–Southern Oscillation (ENSO) conditions during the study period. ENSO influences precipitation and fire conditions differently across locations and seasons (Mataveli et al., 2021). Therefore, ONI should not be interpreted as a direct measure of local drought. The joint visualization is descriptive and does not statistically isolate the effect of ENSO or establish its contribution to annual burned area.

To represent broader institutional conditions in Figure 2A, we plotted the annual V-Dem Rule of Law Index (variable v2x_rule) alongside burned area and ONI (Coppedge, 2026). The rule-of-law measure provides contextual information on institutional conditions rather than directly measuring environmental enforcement. Previous research discusses associations between governance, legal institutions, and environmental outcomes (Börner et al., 2014; Pellegrini and Gerlagh, 2006). We retained the annual indicator values rather than aggregating them by presidential administration, so that their temporal variation remained visible. For visualization, the rule-of-law series was multiplied by a scaling factor calculated from the maximum burned-area value and the maximum rule-of-law value, allowing both series to be displayed with separate vertical axes. The original rule-of-law scale was recovered on the secondary axis. The figure does not estimate the independent effects of governance, climate variability, or political administrations on fire occurrence, and temporal concurrence should not be interpreted as causation.

Spatial and Proportional Distribution of Fire Scars

Raster data containing classified fire-scar sizes for 2015–2023 were obtained from Collection 8 of the MapBiomas Project (MAPBIOMAS, 2026). To summarize the distribution of burned area across patch sizes, we grouped the original raster size classes into three categories: small (≤50 ha), medium (>50–500 ha), and large (>500 ha). For each annual raster, we calculated the frequency of pixels in each original size class and converted pixel counts to burned area by multiplying them by 0.09 ha, the area of a 30 × 30 m pixel. Original classes 1–4 were assigned to the small category, classes 5–6 to the medium category, and classes 7–8 to the large category. For each year, the burned area within each category was summed and divided by the total burned area represented by the analyzed raster classes, yielding the annual percentage of burned area in each size class (Figure 2B).

Patch size describes the spatial extent of a mapped fire scar and does not, by itself, identify ignition source, land use, management practice, or legality. Fire extent can reflect interacting environmental and human influences, including drought, landscape fragmentation, and land-use change (Alencar et al., 2015; Brando et al., 2020). We therefore use the three size classes as descriptive categories rather than proxies for subsistence burning, mechanized clearing, land grabbing, or other specific activities.

Results are displayed as 100% stacked bars to compare the proportional distribution of burned area among size classes across years. The proportions were calculated from the relative contributions of the three aggregated size classes to total mapped burned area in each year. This visualization does not constitute a statistical test of differences in ignition mechanisms or their causes.

Data and code availability

The R scripts and input datasets used to generate Figures 1 and 2, including the MapBiomas burned-area time series and ONI data, are available at https://github.com/biodiva-lab/Political_polarization_environmental_protection_LACA.git.

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